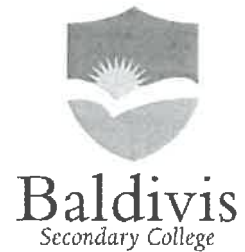


Year 10
Pathway 1

Mini Test 1
Algebra



Name: Soluhons

Score: /33 37
Total Time: 35 minutes
45

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Calculators may be used. 1 side A4 notes (handwritten) allowed.

1) Expand the following expressions

(14 marks)

a) $3(2x + 1)$ $6x + 3$ ✓

b) $-2x(x - 3)$
 $-2x^2 + 6x$ ✓

c) $(x + 8)(x + 1)$
 $x^2 + 9x + 8$ ✓✓

d) $(x + 3)(x - 2)$
 $x^2 + x - 6$ ✓✓

e) $(a - 3)^2$
 $a^2 - 6a + 9$ ✓✓ $a^2 - 9$ or $a^2 - 6a + 9$ 100 marks

f) $(3p - 2)(5p + 11)$
 $15p^2 - 10p + 33p - 22$ ✓✓
 $= 15p^2 + 23p - 22$

g) $(3x + 3)^2 - 2(x + 1)$
 $9x^2 + 18x + 9 - 2x - 2$ ✓✓
 $= 9x^2 + 16x + 7$ ✓

2) Fully factorise the following expressions

(19 marks)

23

a) $5x^2 - 15x$

$5x(x-3)$

1/2 each factor

b) $p(p+7) - 5(p+7)$

$(p-5)(p+7)$

c) $x^2 + 11x + 28$

$(x+7)(x+4)$

d) $x^2 - 9x - 10$

$(x-10)(x+1)$

e) $y^2 - 49$

$(y-7)(y+7)$

f) $3x^2 - 24x + 45$

$= 3(x^2 - 8x + 15)$
 $= 3(x-5)(x-3)$

g) $8 - 2a^2$

$= 2(4 - a^2)$

$= 2(2-a)(2+a)$

h) $2x^2 + 9x + 9 = 2x^2 + 6x + 3x + 9$

~~$(2x+3)(x+3)$~~ $= 2x(x+3) + 3(x+3)$
 $= (2x+3)(x+3)$

i) $4x^2 + 4x - 15$

$= 4x^2 + 10x - 6x - 15$

$= 2x(2x+5) - 3(2x+5)$

$= (2x-3)(2x+5)$

End of Test